

Specific chromosomal abnormalities in malignant human gliomas.

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Karyotypic analysis of 54 malignant human gliomas (5 anaplastic astrocytomas, 43 glioblastoma multiformes, 3 gliosarcomas, 2 giant cell glioblastomas, 1 anaplastic mixed glioma) has demonstrated that 12 tumors contained normal stemlines or only lacked one sex chromosome. The 42 tumors with abnormal karyotypes included 38 tumors which could be completely analyzed. Six of these 38 cases had near-triploid or near-tetraploid stemlines and 32 had near-diploid stemlines. Statistically significant numerical deviations in the near-diploid group were gains of chromosome 7 (26 of 32; P less than 0.001), and losses of chromosome 10 (19 of 32; P less than 0.001). Double minutes occurred in 18 of 32 near diploid tumors. The distribution of structural abnormalities was analyzed statistically by comparing the incidence of breakpoint in each chromosomal arm to the expected value based on chromosomal arm length. This analysis demonstrated that structural abnormalities of 9p and 19q were significant statistically (P less than 0.005 and $P = 0.02$, respectively). Although chromosome 1, 6p, the centromeric region of chromosome 11, 13q, and 15q were also frequently involved in structural abnormalities, the incidence of these breaks did not reach statistical significance. This demonstration of specific chromosomal abnormalities in near-diploid gliomas provides the basis for the investigation of genes which may be quantitatively or qualitatively altered in these neoplasms.

Comparative genomic hybridization reveals recurrent enhancements on chromosome 20 and in one case combined amplification sites on 15q24q26 and 20p11p12 in glioblastomas.

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We examined homogenized tissue samples of biopsies from 19 astrocytomas of different grades for genetic imbalances using comparative genomic hybridization (CGH): three astrocytomas grade II, and 16 astrocytomas grade IV (glioblastoma multiforme), one of the glioblastomas representing the recurrence of a benign oligoastrocytoma. In two of three cases of astrocytoma grade II, a gain of chromosome 7 was found. The alterations in the glioblastomas were complex, and most frequently showed the characteristic gain of chromosome 7 and loss of chromosome 10. The single analyzed case of recurrence of an oligoastrocytoma was characterized by a unique CGH pattern. This tumor showed two distinct alterations: apart from an amplification on 15q24q26, we found a distinct amplification of a small region on 20p11.2p12, which has not been previously described in brain tumors. Partial or complete gains of chromosome 20 arose in six other tumors; we conclude that chromosome 20 in particular 20p11.2p12, may harbor relevant genes for glioma progression.

Most human non-NCIC glioblastoma subtypes evolve from a common proneural-like precursor glioma.

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To understand the relationships between the non-NCIC glioblastoma (GBM) subgroups, we performed mathematical modeling to predict the temporal sequence of driver events during tumorigenesis. The most common order of evolutionary events is 1) chromosome (chr) 7 gain and chr10 loss, followed by 2)

CDKN2A loss and/or TP53 mutation, and 3) alterations canonical for specific subtypes. We then developed a computational methodology to identify drivers of broad copy number changes, identifying PDGFA (chr7) and PTEN (chr10) as driving initial nondisjunction events. These predictions were validated using mouse modeling, showing that PDGFA is sufficient to induce proneural-like gliomas and that additional NF1 loss converts proneural to the mesenchymal subtype. Our findings suggest that most non-GCIMP mesenchymal GBMs arise as, and evolve from, a proneural-like precursor.

Chromosome 7 gain and DNA hypermethylation at the HOXA10 locus are associated with expression of a stem cell related HOX-signature in glioblastoma.

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HOX genes are a family of developmental genes that are expressed neither in the developing forebrain nor in the normal brain. Aberrant expression of a HOX-gene dominated stem-cell signature in glioblastoma has been linked with increased resistance to chemo-radiotherapy and sustained proliferation of glioma initiating cells. Here we describe the epigenetic and genetic alterations and their interactions associated with the expression of this signature in glioblastoma. We observe prominent hypermethylation of the HOXA locus 7p15.2 in glioblastoma in contrast to non-tumoral brain. Hypermethylation is associated with a gain of chromosome 7, a hallmark of glioblastoma, and may compensate for tumor-driven enhanced gene dosage as a rescue mechanism by preventing undue gene expression. We identify the CpG island of the HOXA10 alternative promoter that appears to escape hypermethylation in the HOX-high glioblastoma. An additive effect of gene copy gain at 7p15.2 and DNA methylation at key regulatory CpGs in HOXA10 is significantly associated with HOX-signature expression. Additionally, we show concordance between methylation status and presence of active or inactive chromatin marks in glioblastoma-derived spheres that are HOX-high or HOX-low, respectively. Based on these findings, we propose co-evolution and interaction between gene copy gain, associated with a gain of chromosome 7, and additional epigenetic alterations as key mechanisms triggering a coordinated, but inappropriate, HOX transcriptional program in glioblastoma.

Increased HOXA5 expression provides a selective advantage for gain of whole chromosome 7 in IDH wild-type glioblastoma.

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Glioblastoma is the most frequently occurring and invariably fatal primary brain tumor in adults. The vast majority of glioblastomas is characterized by chromosomal copy number alterations, including gain of whole chromosome 7 and loss of whole chromosome 10. Gain of whole chromosome 7 is an early event in gliomagenesis that occurs in proneural-like precursor cells, which give rise to all isocitrate dehydrogenase (IDH) wild-type glioblastoma transcriptional subtypes. Platelet-derived growth factor A (PDGFA) is one gene on chromosome 7 known to drive gliomagenesis, but, given its location near the end of 7p, there are likely several other genes located along chromosome 7 that select for its increased whole-chromosome copy number within glioblastoma cells. To identify other potential genes that could select for gain of whole chromosome 7, we developed an unbiased bioinformatics approach that identified homeobox A5 (HOXA5) as a gene whose expression correlated with gain of chromosome 7 and a more aggressive phenotype of the resulting glioma. High expression of HOXA5 in glioblastoma was

associated with a proneural gene expression pattern and decreased overall survival in both human proneural and PDGF-driven mouse glioblastoma. Furthermore, HOXA5 overexpression promoted cellular proliferation and potentiated radioresistance. We also found enrichment of HOXA5 expression in recurrent human and mouse glioblastoma at first recurrence after radiotherapy. Overall, this study implicates HOXA5 as a chromosome 7-associated gene-level locus that promotes selection for gain of whole chromosome 7 and an aggressive phenotype in glioblastoma.

Distribution of EGFR amplification, combined chromosome 7 gain and chromosome 10 loss, and TERT promoter mutation in brain tumors and their potential for the reclassification of IDHwt astrocytoma to glioblastoma.

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EGFR amplification (EGFRamp), the combination of gain of chromosome 7 and loss of chromosome 10 (7+/10-), and TERT promoter mutation (pTERTmut) are alterations frequently observed in adult IDH-wild-type (IDHwt) glioblastoma (GBM). In the absence of endothelial proliferation and/or necrosis, these alterations currently are considered to serve as a surrogate for upgrading IDHwt diffuse or anaplastic astrocytoma to GBM. Here, we set out to determine the distribution of EGFRamp, 7+/10-, and pTERTmut by analyzing high-resolution copy-number profiles and next-generation sequencing data of primary brain tumors. In addition, we addressed the question whether combinations of partial gains on chromosome 7 and partial losses on chromosome 10 exhibited a diagnostic and prognostic value similar to that of complete 7+/10-. Several such combinations proved relevant and were combined as the 7/10 signature. Our results demonstrate that EGFRamp and the 7/10 signature are closely associated with IDHwt GBM. In contrast, pTERTmut is less specific for IDHwt GBM. We conclude that, in the absence of endothelial proliferation and/or necrosis, the detection of EGFRamp is a very strong surrogate marker for the diagnosis of GBM in IDHwt diffuse astrocytic tumors. The 7/10 signature is also a strong surrogate marker. However, care should be taken to exclude pleomorphic xanthoastrocytoma. pTERTmut is less restricted to this entity and needs companion analysis by other molecular markers to serve as a surrogate for diagnosing IDHwt GBM. A combination of any two of EGFRamp, the 7/10 signature and pTERTmut, is highly specific for IDHwt GBM and the combination of all three alterations is frequent and exclusively seen in IDHwt GBM.

Copy number gain of FAM131B-AS2 promotes the progression of glioblastoma by mitigating replication stress.

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Glioblastoma (GBM) is characterized by chromosome 7 copy number gains, notably 7q34, potentially contributing to therapeutic resistance, yet the underlying oncogenes have not been fully characterized. Pertinently, the significance of long noncoding RNAs (lncRNAs) in this context has gained attention, necessitating further exploration. FAM131B-AS2 was quantified in GBM samples and cells using qPCR. Overexpression and knockdown of FAM131B-AS2 in GBM cells were used to study its functions in vivo and in vitro. The mechanisms of FAM131B-AS2 were studied using RNA-seq, qPCR, Western blotting, RNA pull-down, coimmunoprecipitation assays, and mass spectrometry analysis. The phenotypic changes that resulted from FAM131B-AS2 variation were evaluated through CCK8 assay, EdU assay, comet assay, and

immunofluorescence. Our analysis of 149 primary GBM patients identified FAM131B-AS2, a lncRNA located in the 7q34 region, whose upregulation predicts poor survival. Mechanistically, FAM131B-AS2 is a crucial regulator of the replication stress response, stabilizing replication protein A1 through recruitment of ubiquitin-specific peptidase 7 and activating the ataxia telangiectasia and rad3-related protein kinase pathway to protect single-stranded DNA from breakage. Furthermore, FAM131B-AS2 overexpression inhibited CD8+ T-cell infiltration, while FAM131B-AS2 inhibition activated the cGAS-STING pathway, increasing lymphocyte infiltration and improving the response to immune checkpoint inhibitors. FAM131B-AS2 emerges as a promising indicator for adjuvant therapy response and could also be a viable candidate for combined immunotherapies against GBMs.

Monosomy of chromosome 10 associated with dysregulation of epidermal growth factor signaling in glioblastomas.

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Glioblastomas--uniformly fatal brain tumors--often have both monosomy of chromosome 10 and gains of the epidermal growth factor receptor (EGFR) gene locus on chromosome 7, an association for which the mechanism is poorly understood. To assess whether coselection of EGFR gains on 7p12 and monosomy 10 in glioblastomas promotes tumorigenic epidermal growth factor (EGF) signaling through loss of the annexin A7 (ANXA7) gene on 10q21.1-q21.2 and whether ANXA7 acts as a tumor suppressor gene by regulating EGFR in glioblastomas. Multidimensional analysis of gene, coding sequence, promoter methylation, messenger RNA (mRNA) transcript, protein data for ANXA7 (and EGFR), and clinical patient data profiles of 543 high-grade gliomas from US medical centers and The Cancer Genome Atlas pilot project (made public 2006-2008; and unpublished, tumors collected 2001-2008). Functional analyses using LN229 and U87 glioblastoma cells. Associations among ANXA7 gene dosage, coding sequence, promoter methylation, mRNA transcript, and protein expression. Effect of ANXA7 haploinsufficiency on EGFR signaling and patient survival. Joint effects of loss of ANXA7 and gain of EGFR expression on tumorigenesis. Heterozygous ANXA7 gene deletion is associated with significant loss of ANXA7 mRNA transcript expression ($P = 1 \times 10^{-15}$; linear regression) and a reduction (mean [SEM]) of 91.5% (2.3%) of ANXA7 protein expression compared with ANXA7 wild-type glioblastomas ($P = .004$; unpaired t test). ANXA7 loss of function stabilizes the EGFR protein (72%-744% increase in EGFR protein abundance) and augments EGFR transforming signaling in glioblastoma cells. ANXA7 haploinsufficiency doubles tumorigenic potential of glioblastoma cells, and combined ANXA7 knockdown and EGFR overexpression promotes tumorigenicity synergistically. The heterozygous loss of ANXA7 in approximately 75% of glioblastomas in the The Cancer Genome Atlas plus infrequency of ANXA7 mutation (approximately 6% of tumors) indicates its role as a haploinsufficiency gene. ANXA7 mRNA transcript expression, dichotomized at the median, associates with patient survival in 191 glioblastomas (log-rank $P = .008$; hazard ratio [HR], 0.667; 95% confidence interval [CI], 0.493-0.902; 46.9 vs 74.8 deaths/100 person-years for high vs low ANXA7 mRNA expression) and with a separate group of 180 high-grade gliomas (log-rank $P = .00003$; HR, 0.476; 95% CI, 0.333-0.680; 21.8 vs 50.0 deaths/100 person-years for high vs low ANXA7 mRNA expression). Deletion of the ANXA7 gene associates with poor patient survival in 189 glioblastomas (log-rank $P = .042$; HR, 0.686; 95% CI, 0.476-0.989; 54.0 vs 80.1 deaths/100 person-years for wild-type ANXA7 vs ANXA7 deletion). Haploinsufficiency of the tumor suppressor ANXA7 due to monosomy of chromosome 10 provides a clinically relevant mechanism to augment EGFR signaling in glioblastomas beyond that resulting from amplification of the EGFR gene.

Array comparative genomic hybridization identifies genetic subgroups in grade 4 human astrocytoma.

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Alterations of DNA copy number are believed to be important indicators of tumor progression in human astrocytoma. We used an array of bacterial artificial chromosomes to map relative DNA copy number in 50 primary glioblastoma multiforme tumors at approximately 1.4-Mb resolution. We identified 33 candidate sites for amplification and homozygous deletion in these tumors. We identified three major genetic subgroups within these glioblastoma multiforme tumors: tumors with chromosome 7 gain and chromosome 10 loss, tumors with only chromosome 10 loss in the absence of chromosome 7 gain, and tumors without copy number change in chromosomes 7 or 10. The significance of these genetic groups to therapeutics needs further study.

DNA copy number analysis of Grade II-III and Grade IV gliomas reveals differences in molecular ontogeny including chromothripsis associated with IDH mutation status.

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Isocitrate dehydrogenase (IDH) mutation status and grade define subgroups of diffuse gliomas differing based on age, tumor location, presentation, and prognosis. While some biologic differences between IDH mutated (IDH (mut)) and wild-type (IDH (wt)) gliomas are clear, the distinct alterations associated with progression of the two subtypes to glioblastoma (GBM, Grade IV) have not been well described. We analyzed copy number alterations (CNAs) across grades (Grade II-III and GBM) in both IDH (mut) and IDH (wt) infiltrating gliomas using molecular inversion probe arrays. Ninety four patient samples were divided into four groups: Grade II-III IDH (wt) (n = 17), Grade II-III IDH (mut) (n = 28), GBM IDH (wt) (n = 25), and GBM IDH (mut) (n = 24). We validated prior observations that IDH (wt) GBM have a high frequency of chromosome 7 gain (including EGFR) and chromosome 10 loss (including PTEN) compared with IDH (mut) GBM. Hierarchical clustering of IDH (mut) gliomas demonstrated distinct CNA patterns distinguishing lower grade gliomas versus GBM. However, similar hierarchical clustering of IDH (wt) gliomas demonstrated no CNA distinction between lower grade glioma and GBM. Functional analyses showed that IDH (wt) gliomas had more chromosome gains in regions containing receptor tyrosine kinase pathways. In contrast, IDH (mut) gliomas more commonly demonstrated amplification of cyclins and cyclin dependent kinase genes. One of the most common alterations associated with transformation of lower grade to GBM IDH (mut) gliomas was the loss of chromosomal regions surrounding PTEN. IDH (mut) GBM tumors demonstrated significantly higher levels of overall CNAs compared to lower grade IDH (mut) tumors and all grades of IDH (wt) tumors, and IDH (mut) GBMs also demonstrated significant increase in incidence of chromothripsis. Taken together, these analyses demonstrate distinct molecular ontogeny between IDH (wt) and IDH (mut) gliomas. Our data also support the novel findings that malignant progression of IDH (mut) gliomas to GBM involves increased genomic instability and genomic catastrophe, while IDH (wt) lower grade tumors are virtually identical to GBMs at the level of DNA copy number alterations.